Monoids in Practice

A monoid is a set with an associative operation and an identity element. Addition over the integers, with zero, is the classic example.

Associativity means the grouping does not matter: (a <> b) <> c equals a <> (b <> c). That is what lets us split work into chunks and combine the results in any grouping.

The identity element leaves values unchanged: mempty <> a equals a. It gives an empty fold somewhere to start.

Everyday monoids

Strings under concatenation, with the empty string as the identity.

Lists under append, with the empty list.

Booleans under logical and, with true; or under logical or, with false.

Maps under union, when the values themselves form a monoid.

Folding with monoids

foldMap turns every element into a monoid and combines the results. Because the operation is associative, the combining can run in parallel.

Many statistics are monoids too: counts, sums, minimums and maximums can all be merged from partial results.